

Module 3: Algebraic Structures

Module Summary

- Semigroups and Monoids
- Groups
- Subgroups
- Lagrange's Theorem
- Homomorphism
- Properties-Group Codes

Let S be any non empty set. Then, a mapping

$$f: S \times S \times \dots \times S \rightarrow S \text{ (or) } f: S^n \rightarrow S$$

is called an n -ary operation and ' n ' called the order of the operation.

Unary operation: $n=1, f: S \rightarrow S$, then this mapping is called the unary operation.

Binary operation: If S is a non-empty set, a binary operation on S is a mapping $f: S \times S \rightarrow S$.

Properties

- **Closure property:** A binary operation $*$: $S \times S \rightarrow S$ is said to be closed if for all $a, b \in S$, $a * b \in S$.
- **Commutative:** Let $*$ be a binary operation on a set A .
The operation $*$ is said to be commutative in A if
 $a * b = b * a$ for all a, b in A
- **Associativity:** Let $*$ be a binary operation on a set A .
The operation $*$ is said to be associative in A if
 $(a * b) * c = a * (b * c)$ for all a, b, c in A
- **Identity:** For an algebraic system $(A, *)$, an element 'e' in A is said to be an identity element of A if
 $a * e = e * a = a$ for all $a \in A$.
- **Note:** For an algebraic system $(A, *)$, the identity element, if exists, is unique.

- **Inverse:** Let $(A, *)$ be an algebraic system with identity 'e'. Let a be an element in A . An element b is said to be inverse of a if

$$a * b = b * a = e$$

- **Distributive properties:**

$$a * (b . c) = (a * b) . (a * c) \text{ (left distributive law)}$$

$$(b . c) * a = (b * a) . (c * a) \text{ (right distributive law)}$$

for all $a, b, c \in S$

- **Cancellation properties:**

$$a * b = a * c \Rightarrow b = c \text{ [left cancellation Law]}$$

$$b * a = c * a \Rightarrow b = c, \text{ [right cancellation Law]}$$

for all $a, b, c \in S$

- **Algebraic System:** A set 'A' with one or more binary(closed) operations defined on it is called an algebraic system.

Ex: $(\mathbb{N}, +)$, $(\mathbb{Z}, +, -)$, $(\mathbb{R}, +, \cdot, -)$ are algebraic systems.

Semi group

- **Semi Group:** An algebraic system $(A, *)$ is said to be a semi group if
 1. $*$ is closed operation on A .
 2. $*$ is an associative operation, for all a, b, c in A .
- Ex. $(\mathbb{N}, +)$ is a semi group.
- Ex. $(\mathbb{N}, \cdot)$ is a semi group.
- Ex. $(\mathbb{N}, -)$ is not a semi group.

- **Monoid:** An algebraic system $(A, *)$ is said to be a **monoid** if the following conditions are satisfied.
 - 1) $*$ is a closed operation in A .
 - 2) $*$ is an associative operation in A .
 - 3) There is an identity in A .

Monoid

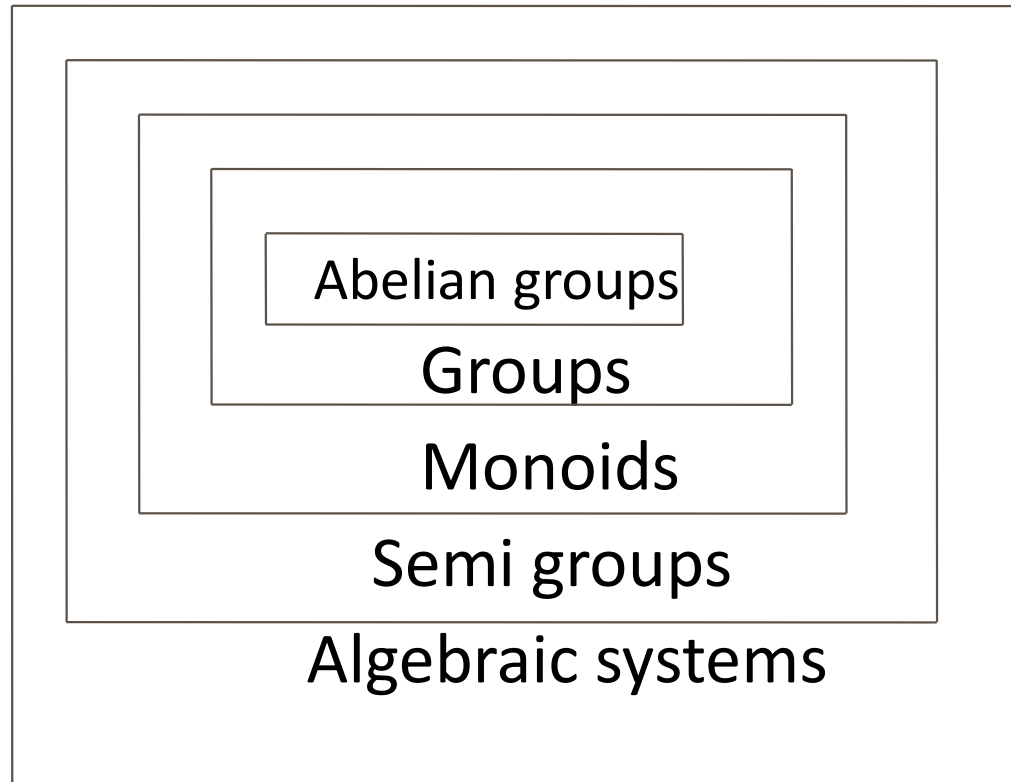
- Ex. Show that the set 'N' is a monoid with respect to multiplication.
 - Solution: Here, $N = \{1, 2, 3, 4, \dots\}$
 1. Closure property: We know that product of two natural numbers is again a natural number.
i.e., $a.b = b.a$ for all $a, b \in N$
 $\therefore$ Multiplication is a closed operation.
 2. Associativity: Multiplication of natural numbers is associative.
i.e., $(a.b).c = a.(b.c)$ for all $a, b, c \in N$
 3. Identity: We have, $1 \in N$ such that
 $a.1 = 1.a = a$ for all $a \in N$.
 $\therefore$ Identity element exists, and 1 is the identity element.
- Hence, N is a monoid with respect to multiplication.

Group

- **Group:** An algebraic system $(G, *)$ is said to be a **group** if the following conditions are satisfied.
 - 1) $*$ is a closed operation.
 - 2) $*$ is an associative operation.
 - 3) There is an identity in G .
 - 4) Every element in G has inverse in G .

- **Abelian group (Commutative group):** A group $(G, *)$ is said to be ***abelian*** (or ***commutative***) if
$$a * b = b * a \quad \text{for all } a, b \in G.$$

Algebraic systems



Theorem

- In a Group $(G, *)$ the following properties hold good

1. Identity element is unique.

2. Inverse of an element is unique.

3. Cancellation laws hold good

$$a * b = a * c \Rightarrow b = c \quad (\text{left cancellation law})$$

$$a * c = b * c \Rightarrow a = b \quad (\text{Right cancellation law})$$

4. $(a * b)^{-1} = b^{-1} * a^{-1}$

- In a group, the identity element is its own inverse.

- **Order of a group** : The number of elements in a group is called order of the group.

- **Finite group**: If the order of a group G is finite, then G is called a finite group.

Ex. Show that, the set of all integers is a group with respect to addition.

■ Solution: Let Z = set of all integers.

Let a, b, c are any three elements of Z .

1. Closure property : We know that, Sum of two integers is again an integer.

i.e., $a + b \in Z$ for all $a, b \in Z$

2. Associativity: We know that addition of integers is associative.

i.e., $(a+b)+c = a+(b+c)$ for all $a, b, c \in Z$.

3. Identity: We have $0 \in Z$ and $a + 0 = a$ for all $a \in Z$.

$\therefore$ Identity element exists, and '0' is the identity element.

4. Inverse: To each $a \in Z$, we have $-a \in Z$ such that

$$a + (-a) = 0$$

Each element in Z has an inverse.

Contd.,

- 5. Commutativity: We know that addition of integers is commutative.
i.e., $a + b = b + a$ for all $a, b \in \mathbb{Z}$.
Hence, $(\mathbb{Z}, +)$ is an abelian group.

Ex. Show that set of all non zero real numbers is a group with respect to multiplication .

■ Solution: Let R^* = set of all non zero real numbers.

Let a, b, c are any three elements of R^* .

1. Closure property : We know that, product of two nonzero real numbers is again a nonzero real number .

i.e., $a \cdot b \in R^*$ for all $a, b \in R^*$.

2. Associativity: We know that multiplication of real numbers is associative.

i.e., $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ for all $a, b, c \in R^*$.

3. Identity: We have $1 \in R^*$ and $a \cdot 1 = a$ for all $a \in R^*$.

$\therefore$ Identity element exists, and '1' is the identity element.

4. Inverse: To each $a \in R^*$, we have $1/a \in R^*$ such that

$a \cdot (1/a) = 1$ i.e., Each element in R^* has an inverse.

Contd.,

- 5.Commutativity: We know that multiplication of real numbers is commutative.
i.e., $a \cdot b = b \cdot a$ for all $a, b \in \mathbb{R}^*$.
Hence, $(\mathbb{R}^*, \cdot)$ is an abelian group.
- Ex: Show that set of all real numbers 'R' is not a group with respect to multiplication.
- Solution: We have $0 \in \mathbb{R}$.
The multiplicative inverse of 0 does not exist.
Hence. $\mathbb{R}$ is not a group.

Example

- Ex. Let $(Z, *)$ be an algebraic structure, where Z is the set of integers and the operation $*$ is defined by $n * m = \text{maximum of } (n, m)$. Show that $(Z, *)$ is a semi group.
Is $(Z, *)$ a monoid ?. Justify your answer.

- Solution: Let a, b and c are any three integers.

Closure property: Now, $a * b = \text{maximum of } (a, b) \in Z$ for all $a, b \in Z$

Associativity : $(a * b) * c = \text{maximum of } \{a, b, c\} = a * (b * c)$

$\therefore (Z, *)$ is a semi group.

Identity : There is no integer x such that

$$a * x = \text{maximum of } (a, x) = a \quad \text{for all } a \in Z$$

$\therefore$ Identity element does not exist. Hence, $(Z, *)$ is not a monoid.

Ex. If M is set of all non singular matrices of order ' $n \times n$ '.
then show that M is a group w.r.t. matrix multiplication.
Is $(M, *)$ an abelian group?. Justify your answer.

■ Solution: Let $A, B, C \in M$.

1. Closure property : Product of two non singular matrices is again a non singular matrix, because

$$|AB| = |A| \cdot |B| \neq 0 \quad (\text{Since, } A \text{ and } B \text{ are nonsingular})$$

i.e., $AB \in M$ for all $A, B \in M$.

2. Associativity: Matrix multiplication is associative.

$$\text{i.e., } (AB)C = A(BC) \quad \text{for all } A, B, C \in M.$$

3. Identity: We have $I_n \in M$ and $A I_n = A$ for all $A \in M$.

$\therefore$ Identity element exists, and ' I_n ' is the identity element.

4. Inverse: To each $A \in M$, we have $A^{-1} \in M$ such that

$$A A^{-1} = I_n \quad \text{i.e., Each element in } M \text{ has an inverse.}$$

Contd.,

- $\therefore$ M is a group w.r.t. matrix multiplication.

We know that, matrix multiplication is not commutative.

Hence, M is not an abelian group.

Ex. Show that the set of all positive rational numbers forms an abelian group under the composition $*$ defined by
 $a * b = (ab)/2$.

■ Solution: Let A = set of all positive rational numbers.

Let a, b, c be any three elements of A .

1. Closure property: We know that, Product of two positive rational numbers is again a rational number.

i.e., $a * b \in A$ for all $a, b \in A$.

2. Associativity: $(a * b) * c = (ab/2) * c = (abc) / 4$
 $a * (b * c) = a * (bc/2) = (abc) / 4$

3. Identity: Let e be the identity element.

We have $a * e = (a e)/2 \dots(1)$, By the definition of $*$
again, $a * e = a \dots(2)$, Since e is the identity.

From (1) and (2), $(a e)/2 = a \Rightarrow e = 2$ and $2 \in A$.

$\therefore$ Identity element exists, and '2' is the identity element in A .

Contd.,

■ 4. Inverse: Let $a \in A$

let us suppose b is inverse of a .

Now, $a * b = (a b)/2 \dots(1)$ (By definition of inverse.)

Again, $a * b = e = 2 \dots(2)$ (By definition of inverse)

From (1) and (2), it follows that

$$(a b)/2 = 2$$

$$\Rightarrow b = (4 / a) \in A$$

$\therefore (A, *)$ is a group.

Commutativity: $a * b = (ab/2) = (ba/2) = b * a$

Hence, $(A, *)$ is an abelian group.

Ex: Check if the followings are group or not

The operation in Z_n with respect to addition

The operation in U_n with respect to multiplication

Theorem

- Ex. In a group $(G, *)$, Prove that the identity element is unique.

- Proof:

Let e_1 and e_2 are two identity elements in G .

Now, $e_1 * e_2 = e_1$...(1) (since e_2 is the identity)

Again, $e_1 * e_2 = e_2$...(2) (since e_1 is the identity)

From (1) and (2), we have $e_1 = e_2$

Hence, identity element in a group is unique.

Theorem

■ Ex. In a group $(G, *)$, Prove that the inverse of any element is unique.

■ Proof:

Let $a, b, c \in G$ and e is the identity in G .

Let us suppose, Both b and c are inverse elements of a .

Now, $a * b = e$...(1) (Since, b is inverse of a)

Again, $a * c = e$...(2) (Since, c is also inverse of a)

From (1) and (2), we have

$$a * b = a * c$$

$\Rightarrow b = c$ (By left cancellation law)

In a group, the inverse of any element is unique.

■ Ex. In a group $(G, *)$, Prove that
 $(a * b)^{-1} = b^{-1} * a^{-1}$ for all $a, b \in G$.

■ Proof :

Consider,

$$\begin{aligned} & (a * b) * (b^{-1} * a^{-1}) \\ &= (a * (b * b^{-1}) * a^{-1}) && \text{(By associative property).} \\ &= (a * e * a^{-1}) && \text{(By inverse property)} \\ &= (a * a^{-1}) && \text{(Since, e is identity)} \\ &= e && \text{(By inverse property)} \end{aligned}$$

Similarly, we can show that

$$(b^{-1} * a^{-1}) * (a * b) = e$$

$$\text{Hence, } (a * b)^{-1} = b^{-1} * a^{-1}.$$

■ Ex. If $(G, *)$ is a group and $a \in G$ such that $a * a = a$, then show that $a = e$, where e is identity element in G .

■ Proof: Given that, $a * a = a$
 $\Rightarrow a * a = a * e$ (Since, e is identity in G)
 $\Rightarrow a = e$ (By left cancellation law)

Hence, the result follows.

■ Ex. If every element of a group is its own inverse, then show that the group must be abelian .

■ Proof: Let $(G, *)$ be a group.

Let a and b are any two elements of G .

Consider the identity,

$$(a * b)^{-1} = b^{-1} * a^{-1}$$

$\Rightarrow (a * b) = b * a$ (Since each element of G is its own inverse)

Hence, G is abelian.

Theorem

- Ex. In a group $(G, *)$, if $(a * b)^2 = a^2 * b^2 \forall a, b \in G$ then show that G is abelian group.
- Proof: Given that $(a * b)^2 = a^2 * b^2$
 - $\Rightarrow (a * b) * (a * b) = (a * a) * (b * b)$
 - $\Rightarrow a * (b * a) * b = a * (a * b) * b$ (By associative law)
 - $\Rightarrow (b * a) * b = (a * b) * b$ (By left cancellation law)
 - $\Rightarrow (b * a) = (a * b)$ (By right cancellation law)

Hence, G is abelian group.

Note: $a^2 = a * a$
 $a^3 = a * a * a$ etc.

Sub groups

- Def. A non empty sub set H of a group $(G, *)$ is a sub group of G , if $(H, *)$ is a group.

Note: For any group $\{G, *\}$, $\{e, *\}$ and $(G, *)$ are trivial sub groups.

- Theorem: A non empty sub set H of a group $(G, *)$ is a sub group of G iff

i) $a * b \in H \quad \forall a, b \in H$

ii) $a^{-1} \in H \quad \forall a \in H$

- Ex. $G = \{1, -1, i, -i\}$ is a group w.r.t multiplication.

$$H_1 = \{1, -1\} \text{ is a subgroup of } G.$$

$$H_2 = \{1\} \text{ is a trivial subgroup of } G.$$

- Ex. $(\mathbb{Z}, +)$ and $(\mathbb{Q}, +)$ are sub groups of the group $(\mathbb{R}, +)$.
- The set of even integer $2\mathbb{Z} = \{2k/k \in \mathbb{Z}\}$ is a subgroup of $(\mathbb{Z}, +)$
- The set of non-negative integers is not a subgroup of $(\mathbb{Z}, +)$, because, except 0, no element have the additive inverse.

- Theorem: A necessary and sufficient condition for a non empty subset H of a group $(G, *)$ to be a sub group is that

$$a \in H, b \in H \Rightarrow a * b^{-1} \in H.$$

- Proof:

Necessary Conditions: Let $(G, *)$ be a group and H is a subgroup of G

$$\text{Let } a, b \in H \Rightarrow b^{-1} \in H \quad (\text{since } H \text{ is a group})$$

$$\Rightarrow a * b^{-1} \in H. \quad (\text{By closure property in } H)$$

Sufficient Conditions: Let H be a non empty set of a group $(G, *)$.

$$\text{Let } a * b^{-1} \in H \quad \forall a, b \in H$$

$$\text{Now, } a * a^{-1} \in H \quad (\text{Taking } b = a)$$

$$\Rightarrow e \in H \quad \text{i.e., identity exists in } H.$$

$$\text{Now, } e \in H, a \in H \Rightarrow e * a^{-1} \in H$$

$$\Rightarrow a^{-1} \in H$$

Contd.,

$\therefore$ Each element of H has inverse in H .

- Further, $a \in H, b \in H \Rightarrow a \in H, b^{-1} \in H$

$$\Rightarrow a * (b^{-1})^{-1} \in H.$$

$$\Rightarrow a * b \in H.$$

$\therefore H$ is closed w.r.t $*$.

- Finally, Let $a, b, c \in H$

$$\Rightarrow a, b, c \in G \text{ (since } H \subseteq G \text{)}$$

$$\Rightarrow (a * b) * c = a * (b * c)$$

$\therefore *$ is associative in H

- Hence, H is a subgroup of G .

Ex. Show that the intersection of two sub groups of a group G is again a sub group of G .

■ Proof:

Let $(G, *)$ be a group.

Let H_1 and H_2 are two sub groups of G .

Let $a, b \in H_1 \cap H_2$.

Now, $a, b \in H_1 \Rightarrow a * b^{-1} \in H_1$ (Since, H_1 is a subgroup of G)

again, $a, b \in H_2 \Rightarrow a * b^{-1} \in H_2$ (Since, H_2 is a subgroup of G)

$\therefore a * b^{-1} \in H_1 \cap H_2$.

Hence, $H_1 \cap H_2$ is a subgroup of G .

Ex. Show that the union of two sub groups of a group G need not be a sub group of G .

■ Proof:

Let G be an additive group of integers.

Let $H_1 = \{ 0, \pm 2, \pm 4, \pm 6, \pm 8, \dots \}$

and $H_2 = \{ 0, \pm 3, \pm 6, \pm 9, \pm 12, \dots \}$

Here, H_1 and H_2 are groups w.r.t addition.

Further, H_1 and H_2 are subsets of G .

$\therefore H_1$ and H_2 are sub groups of G .

$H_1 \cup H_2 = \{ 0, \pm 2, \pm 3, \pm 4, \pm 6, \dots \}$

Here, $H_1 \cup H_2$ is not closed w.r.t addition.

For ex. $2, 3 \in G$

But, $2 + 3 = 5$ and 5 does not belongs to $H_1 \cup H_2$.

Hence, $H_1 \cup H_2$ is not a sub group of G .

Cyclic Group

Definition:

A group $(G,*)$ is said to be cyclic if there exists $a \in G$ such that any $x \in G$ can be written as either $x = a^n$ or $x = na$, where n is some integer. Here the element ' a ' is called the 'generator' of the cyclic group G . That is the cyclic group generated by ' a ' and we denote it by $G = \langle a \rangle$

Theorem 1:

Every cyclic group is abelian.

Proof:

Let $(G,*)$ be a cyclic group generated by an element $a \in G$. i.e $G = \langle a \rangle$. Then for any two elements $x, y \in G$, we have $x = a^n, y = a^m$, where m, n are integers.

$$\begin{aligned} x * y &= a^n * a^m = a^{n+m} = a^{m+n} \\ &= a^m * a^n \\ &= y * x \end{aligned}$$

$(G,*)$ is abelian.

Theorem 2:

If 'a' is a generator of a cyclic group G , then a^{-1} is also a generator of G (or). If a is an element of group G , then $\langle a \rangle = \langle a^{-1} \rangle$.

Proof. Let $G = \langle a \rangle$ be a cyclic group generated by 'a'. Then $a^r \in G$, where r is some integer we can write $a^r = (a^{-1})^{-r}$, since $-r$ is also some integer. Each element of G is generated by a^{-1} .

a^{-1} is also a generator of G . $\langle G \rangle = \langle a \rangle = \langle a^{-1} \rangle$.

Theorem 3:

Prove that the group $G = \{1, -1, i, -i\}$ is cyclic and find its generators.

Solution. We can write $1 = (i)^4, -1 = (i)^2, i = (i)^1, -i = (i)^3$

i.e, All the elements of G can be expressed as integral powers of the element i.e.

G is a cyclic group generated by i . Since 'i' is the generator of G , $(i)^{-1}$ is also a generator of G . Hence G is a cyclic group and its generators are i and $(i)^{-1} = 1/i = -i$

Theorem 4:

Every subgroup of a cyclic group is cyclic

Theorem 5:

Let $(G,*)$ be a finite group generated by an element $a \in G$. If G is of order n , that is $O(G) = n$, then $a^n = e$ so that $G = \{a, a^2, \dots a^n = e\}$. Further, n is the least positive integer for which $a^n = e$.

Homomorphism and Isomorphism.

Homomorphism : Let $(G,*)$ and (H,Δ) be any two groups. A mapping $f: G \rightarrow H$ is said to be a homomorphism if $f(a*b) = f(a)\Delta f(b)$, for any $a, b \in G$.

Ex1: Let $g: (Z, +) \rightarrow (Z, +)$ defined by $g(x) = 2x$ for all $x \in Z$

Then for $x, y \in Z$

$$g(x + y) = 2(x + y) = 2x + 2y = g(x) + g(y)$$

$\therefore g$ is a homomorphism.

Ex2: Let $(R^+, \cdot)$ be a group of positive real numbers under multiplication. Let $(R, +)$ be the group of all real numbers under addition.

Define $f: R^+ \rightarrow R$ by $f(x) = \log_{10}^x$. The binary operation for the groups R^+ and R are $\cdot$ and $+$ respectively.

To prove: f is a homomorphism.

i.e. to show that $f(xy) = f(x) + f(y)$ for all $x, y \in R^+$

Now,

$$f(xy) = \log_{10}^{xy} = \log_{10}^x + \log_{10}^y = f(x) + f(y).$$

$\therefore f$ is a homomorphism.

Theorem: Consider the groups $(G_1, *)$ and $(G_2, \oplus)$ with identity elements e_1 and e_2 respectively. If $f : G_1 \rightarrow G_2$ is a group homomorphism, then prove that

a) $f(e_1) = e_2$

b) $f(a^{-1}) = [f(a)]^{-1}$

Proof: a) We have in G_2 ,

$$\begin{aligned} e_2 \oplus f(e_1) &= f(e_1) && \text{(since, } e_2 \text{ is identity in } G_2 \text{)} \\ &= f(e_1 * e_1) && \text{(since, } e_1 \text{ is identity in } G_1 \text{)} \\ &= f(e_1) \oplus f(e_1) && \text{(since } f \text{ is a homomorphism)} \\ e_2 &= f(e_1) && \text{(By right cancellation law)} \end{aligned}$$

b) For any $a \in G_1$, we have

$$f(a) \oplus f(a^{-1}) = f(a * a^{-1}) = f(e_1) = e_2$$

$$\text{and } f(a^{-1}) \oplus f(a) = f(a^{-1} * a) = f(e_1) = e_2$$

$$\therefore f(a^{-1}) \text{ is the inverse of } f(a) \text{ in } G_2 \text{ i.e., } [f(a)]^{-1} = f(a^{-1})$$

Theorem 2: Let f be a homomorphism from G to G' . Then $f(G)$ (the homomorphic image of G into G') is a subgroup of G'

Proof:

Let $f(G) = \{f(x) \mid x \in G\}$

Clearly $f(G)$ is a non-empty subset of G' . Now for any $a', b' \in f(G)$, we have $f(a) = a', f(b) = b'$, for $a, b \in G$.

$$\begin{aligned} a' * (b')^{-1} &= f(a) * f[b]^{-1} \\ &= f(a) * f(b^{-1}) \\ &= f(a * b^{-1}) \end{aligned}$$

But $a * b^{-1} \in G \Rightarrow f(a * b^{-1}) \in f(G) \Rightarrow a' * (b')^{-1} \in f(G)$

i.e., For $a', b' \in f(G) \Rightarrow a' * (b')^{-1} \in f(G)$

$\Rightarrow f(G)$ is a subgroup of G' .

Isomorphism

Definition: A mapping f from a Group $(G, .)$ to a group $(G', *)$ is said to be an **Isomorphism** if

- f is a homomorphism i.e. $f(a . b) = f(a) * f(b)$, for any $a, b \in G$
- f is one-one. (injective) i.e. distinct elements of G have different f -image in G' .
- f is onto (surjective) i.e. every element of G' should be the f -image of some elements in G .

Ex 1: Let $f: (Z, +) \rightarrow (2Z, +)$ defined by $f(x) = 2x$ for all $x \in Z$ prove that f is an isomorphism.

Solution:

Given $f: (Z, +) \rightarrow (2Z, +)$

Defined by $f(x) = 2x$

Now $f(x + y) = 2(x + y) = 2x + 2y = f(x) + f(y)$

for all $x, y \in Z$

Therefore, f is a homomorphism.

Let $f(x) = f(y) \Rightarrow 2x = 2y \Rightarrow x = y$

$\therefore f$ is 1-1

Clearly f is onto.

Hence f is an isomorphism.

Ex 2. Let R be a group of all real numbers under addition and R^+ be a group of all positive real numbers under multiplication. Show that the mapping $f : R \rightarrow R^+$ defined by $f(x) = 2^x$ for all $x \in R$ is an isomorphism.

Solution: Homomorphism property is already shown.

Next, let us prove that f is a Bijection.

For any $a, b \in R$, Let, $f(a) = f(b)$

$$\Rightarrow 2^a = 2^b$$

$$\Rightarrow a = b$$

$\therefore f$ is one-to-one.

Next, take any $c \in R^+$.

Then $\log_2 c \in R$ and $f(\log_2 c) = 2^{\log_2 c} = c$.

$\Rightarrow$ Every element in R^+ has a pre image in R .

i.e., f is onto.

$\therefore f$ is a bijection.

Hence, f is an isomorphism.

Coset

Definition: If G is a group and H is a subgroup of G , then,

- for any $a \in G$, the set $a * H$ defined by $a * H = \{a * h \mid h \in H\}$ is called the **Left coset** of H in G determined by the element $a \in G$.

- for any $a \in G$, the set $H * a$ defined by $H * a = \{h * a \mid h \in H\}$ is called the **Right coset** of H in G determined by the element $a \in G$.

The element $a \in G$ is called the representative element of the left coset $a * H$ and right coset $H * a$

Note The right or left coset of H in G is not empty.

Since $e \in H$, $e * H = H = H * e$, H itself is right as well as left coset. $H * a$, $a * H$ are also subsets of G .

If G is abelian, then $a * H = H * a$ In all the subsequence discussion in this chapter, aH means $a * H$ and Ha means $H * a$

Example.1

Let $G = \{1, a, a^2, a^3\}$ be a group and $H = \{1, a^2\}$ is a subgroup of G under multiplication. Find all the cosets of H .

Solution:

Let us find the right cosets of H in G .

$$H1 = \{1, a^2\} = H$$

$$Ha = \{a, a^3\},$$

$$H a^2 = \{a^2, a^4\} = \{a^2, 1\} = H$$

$$H a^3 = \{a^3, a^5\} = \{a^3, a\} = Ha$$

$$H.1 = H = H a^2 = \{1, a^2\} \text{ and } Ha = H a^3 = \{a, a^3\}$$

are two distinct right cosets of H in G . Similarly we can find the left cosets of H in G .

Example.2 Find the left cosets of $\{[0],[2]\}$ in the group $(Z_4, +_4)$.

Solution Let $Z_4 = \{[0],[1],[2],[3]\}$ be a group and $H = \{[0],[2]\}$ be a subgroup of Z_4 under $+_4$ (addition mode 4).

The left cosets of H are

$$[0] + H = \{[0],[2]\} = H ;$$

$$[1] + H = \{[1],[3]\} ;$$

$$[2] + H = \{[2],[4]\} = \{[2],[0]\} = \{[0],[2]\} = H$$

$$[3] + H = \{[3],[5]\} = \{[3],[1]\} = \{[1],[3]\} = [1] + H.$$

$[0] + H = [2] + H = H$ and $[1] + H = [3] + H$ are the two distinct left cosets of H in Z_4 .

Note: The union of all distinct left cosets of H in G is equal to G .
If H any subgroup of G and $a \in H$, then $Ha = H = aH$.

Theorem: Any two right cosets of H in G are either disjoint or identical.

Lagrange's theorem

The order of each subgroup of a finite group is a divisor of the order of the group.

INTRODUCTION TO CODING THEORY

- The process of communication involves transmitting some information carrying signal(message) that is conveyed by a sender to a receiver.
- Even though the sender may like to have his message received by the receiver without any distortion, it is not possible due to a variety of disturbances(noise) to which the communication channel is subjected.
- Coding theory deals with minimizing the distortions of the conveyed message due to noise and to retrieve the original message to the optimal extent possible from the corrupted message.

ENCODERS AND DECODERS

Encoder:

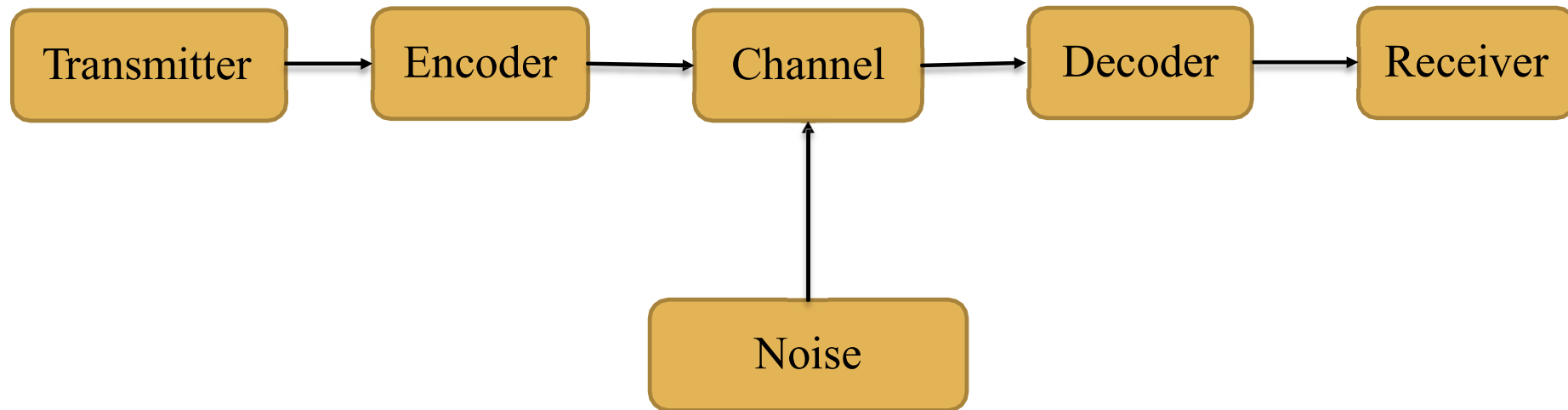
An encoder is a device which transforms the incoming messages in such a way that the presence of noise in the transformed messages is detectable.

Decoder:

A decoder is a device which transforms the encoded message into their original form that can be understood by the receiver.

- So, by using a suitable encoder and decoder, it may be possible to detect the distortions in the messages due to noise in the channel and to correct them.

TYPICAL DATA COMMUNICATION MODEL WITH NOISE



EXPLANATION

- The input message can consist of a sequence of letters, characters or symbol set called alphabet.
- The input message will be transformed by the encoder into a string of characters or symbols of another alphabet in a one-to-one fashion.
- We will discuss a binary channel in which the encoder will transform an input message into a binary string consisting of the symbols 0 and 1.
- Decoding can be seen as the inverse operation of encoding.

GROUP CODE

Definition:

If $B = \{0,1\}$ then $B^n = \{x_1, x_2, x_3, \dots, x_n \mid x_i \in B, i=1,2,3, \dots n\}$ is a group under the binary operation of addition modulo 2 (denoted by $\oplus$), then $(B^n, \oplus)$ is called a **groupcode**.

HAMMING CODES

Definition:

- The codes obtained by introducing additional digits called *parity digits* to the digits in the original message are called **hamming codes**.

In Hamming's single-error detecting code, the last digit is made either 0 or 1 and the remaining digits contain the information part of the message.

If the digit introduced in the last position gives an even number/odd number of 1's in the encoded word, the extra digit is called an even/ odd **parity check**.

EXAMPLE OF PARITY CHECK

Code before parity check	Code after even parity check	Code after odd parity check
000	0000	0001
001	0011	0010
010	0101	0100
011	0110	0111
100	1001	1000
101	1010	1011
110	1100	1101
111	1111	1110

HAMMING DISTANCE

Definition:

- Suppose if x and y represents binary strings $x_1x_2x_3 \dots x_n$ and $y_1y_2y_3 \dots y_n$ then the number of positions in the strings for which $x_i \neq y_i$ is called **Hamming Distance** between x and y .
- Denoted by $H(x,y)$

$$H(x,y) = \text{weight of } x \oplus y = |x \oplus y|$$



The number of 1's in the binary string

BASIC NOTATIONS OF ERRORS CORRECTION USING MATRICES

- Where $m, n \in \mathbb{Z}^+$, ($m < n$) { m should be strictly less than n }, the coding function $e: B^m \rightarrow B^n$ [Where $B = (0,1)$] is given by a $m \times n$ matrix G over B .
- The matrix G is called as generator matrix for the code and is of the form $[I_m | A]$ where I_m is the $m \times m$ unit matrix and A is an $m \times (n-m)$ matrix chosen suitably.

$$H = [A^T | I_{n-m}]$$

- If w is a message $\in B^m$ then $e(w) = wG$ and the code (the set of code words) $c = e(B^m) \subseteq B^n$, where w is a $(1 \times m)$ vector.

EXAMPLE

- Let $w \in B^2$ and G is $\begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix}$
- The words belonging to B^2 are 00, 01, 10, 11. Then we have the code words as:

$$e(00) = [0 \ 0] \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} = [00 \ 000]$$

$$e(10) = [1 \ 0] \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} = [10 \ 110]$$

$$e(01) = [0 \ 1] \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} = [01 \ 011]$$

$$e(11) = [1 \ 1] \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} = [111 \ 01]$$

- Question. Find the code words generated by the encoding function $e: B^m \rightarrow B^n$ with respect to the parity check matrix given by

$$H = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Note: In our discussion, if the encoding function is $e: B^m \rightarrow B^n$, the generator matrix was assumed as an $m \times n$ matrix $G = [I_m | A]$ and the parity check matrix was assumed as an $(n - m) \times m$ matrix $H = [A^T | I_{n-m}]$ and as such there was less number of rows and more number of columns in H . We shall stick to our notation. As per our notation, what is given in this problem is not H , but H^T . However some authors use this notation to denote the parity check matrix.

Solution: Let $e = B^m \rightarrow B^n$ where $m < n$ be the encoding function.

Here $n = 5$ and $m = 2$.

Hence, the generator matrix G is given by

$$G = [I_m | A] = \left[\begin{array}{cc|cc} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{array} \right]$$

Now $B^2 \equiv \{0 \ 0, 0 \ 1, 1 \ 0, 1 \ 1\}$ and $e(w) = w \ G$

$$\therefore e(0 \ 0) = [0 \ 0] \left[\begin{array}{cc|cc} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{array} \right] = [0 \ 0 \ 0 \ 0 \ 0]$$

$$\therefore e(0 \ 1) = [0 \ 1] \left[\begin{array}{cc|cc} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{array} \right] = [0 \ 1 \ 0 \ 1 \ 1]$$

$$\therefore e(1 \ 0) = [1 \ 0] \left[\begin{array}{cc|cc} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{array} \right] = [1 \ 0 \ 0 \ 1 \ 1]$$

$$\therefore e(1 \ 1) = [1 \ 1] \left[\begin{array}{cc|cc} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{array} \right] = [1 \ 1 \ 0 \ 0 \ 0]$$

Hence, the code words generated by H are 0 0 0 0 0, 0 1 0 1 1, 1 0 0 1 1 and 1 1 0 0 0.

- Question. Find the code words generated by the parity check matrix given by

$$H = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

when the encoding function is $e: B^3 \rightarrow B^6$.

Solution:

Let $e : B^m \rightarrow B^n$ where $m < n$ be the encoding function. Here, $m=3$ and $n=6$

$G = [I_m \mid A]$ be the generated matrix, So, we have

$$G = [I_m \mid A] = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{array} \right]$$

Now, $B^3 = \{000, 001, 010, 100, 101, 110, 111\}$

From $e(w) = wG$ we have :

$$\begin{aligned}\therefore \quad e(0 \ 0 \ 0) &= [0 \ 0 \ 0] \cdot G = [0 \ 0 \ 0 \ 0 \ 0 \ 0] \\ e(0 \ 0 \ 1) &= [0 \ 0 \ 1] \cdot G = [0 \ 0 \ 1 \ 0 \ 1 \ 1] \\ e(0 \ 1 \ 0) &= [0 \ 1 \ 0] \cdot G = [0 \ 1 \ 0 \ 1 \ 0 \ 1] \\ e(1 \ 0 \ 0) &= [1 \ 0 \ 0] \cdot G = [1 \ 0 \ 0 \ 1 \ 1 \ 1] \\ e(0 \ 1 \ 1) &= [0 \ 1 \ 1] \cdot G = [0 \ 1 \ 1 \ 1 \ 1 \ 0] \\ e(1 \ 0 \ 1) &= [1 \ 0 \ 1] \cdot G = [1 \ 0 \ 1 \ 1 \ 0 \ 0] \\ e(1 \ 1 \ 0) &= [1 \ 1 \ 0] \cdot G = [1 \ 1 \ 0 \ 0 \ 1 \ 0] \\ e(1 \ 1 \ 1) &= [1 \ 1 \ 1] \cdot G = [1 \ 1 \ 1 \ 0 \ 0 \ 1]\end{aligned}$$

Thus, the code words generated are 0 0 0 0 0 0, 0 0 1 0 1 1,
0 1 0 1 0 1, 1 0 0 1 1 1, 0 1 1 1 1 0, 1 0 1 1 0 0, 1 1 0 0 1 0
and 1 1 1 0 0 1.

PARITY CHECK MATRIX

- Parity check matrix is used for decoding at receiver.
- Parity check matrix is denoted by H
- $H = [A^T \mid I_{n-k}]$
- Where A is same parity bits matrix that is used to construct G(generator matrix) for encode and I is identity matrix.

ERROR DETECTION USING PARITY CHECK MATRIX

Given, r : which is received word

H : unique parity check matrix which corrects the single error in transmission.

Three cases:

$$1. H \cdot r^t = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

There is no error in transition and r is the code word transmitted.

ERROR DETECTION USING PARITY CHECK MATRIX

2. If $H \cdot r^t =$ i th column of H , then we conclude that there is a single error in r during transmission and it has occurred in the i th component of r . Changing the i th component of r , we get the code word c .

$$H \cdot r^t = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

ERROR DETECTION USING PARITY CHECK MATRIX

3. If neither case (i) nor case (ii) occurs we conclude that more than one transmission error have occurred. Detection is possible in this case but correction is not possible.

$H \cdot r^t$ is not equal to any column of H

$$H \cdot r^t = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

PRACTICE QUESTIONS

Q1. Given the generator matrix, $G =$

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

Corresponding to the encoding function, $e: B^3 \rightarrow B^6$, find the corresponding parity check matrix and use it to decode the following received words and hence, to find the original message. Are all words decoded uniquely?

(i) 110101 (ii) 001111 (iii) 111111

PRACTICE QUESTIONS

solution : we assume $G = [I_3 \mid A]$

Then, $H = [A^T \mid I_3]$ $H = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}$

(i) $H \cdot r^T = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{pmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Since it is 0, the received word in this case is the transmitted word itself. Original message is 110.

PRACTICE QUESTIONS

(ii)

$$H \cdot r^T = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{pmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Since $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ is the fifth column of H , the element in the fifth position of r is changed.

The decoded word is 001101 and original message is 001.

PRACTICE QUESTIONS

(iii)

$H \cdot r^T =$

$$\begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{pmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Since, the syndrome is not identical with any column of H , the received word cannot be decoded uniquely.